

BCHAIN-ACCESS Evaluation Instruments

Exact blank questionnaires and walkthrough scripts administered to the expert study panel (N=10)

1. System Usability Scale (SUS)

Each item was scored by the evaluators on a standard 5-point Likert scale: **1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree.**

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

2. Accessibility Questionnaire (WCAG 2.1 AA User Experience)

Scored on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree) to measure practical accessibility:

1. I found the navigation consistent and predictable.
2. I found the instructions and labels easy to understand.
3. I was able to access all features using only the keyboard.
4. I found the colour contrast adequate for reading.
5. I found the error messages helpful and clear.
6. I felt in control while using the system.
7. I would feel comfortable using this system independently.
8. I found the system forgiving if I made a mistake.
9. I understood the information presented to me.
10. I found the system responsive and fast enough.

3. Cognitive Walkthrough Task Scripts

The evaluators performed the following 8 tasks sequentially to evaluate the reference implementation:

Task 1: Biometric Unlock: Authenticate and unlock your health dashboard using the simulated fingerprint/biometric reader on the landing screen.

Task 2: View Record: Locate your 'MRI Brain Scan' record on the dashboard and view its metadata details.

Task 3: Access Verification: Verify the current list of authorized providers for this record to ensure no unauthorized access is active.

Task 4: Access Grant Initiation: Select Dr. Sarah Chen from the provider directory and initiate a 'View Only' access grant for your MRI Brain Scan record.

Task 5: Review Cooldown: Review the pending access grant details and inspect the active 60-second safety cooldown window.

Task 6: Access Confirmation: Wait for the 60-second safety countdown to expire, then confirm the access grant to finalize it on-chain.

Task 7: Access Revocation: Immediately revoke Dr. Sarah Chen's access permissions to verify that access is cut off with no pending delay.

Task 8: Help & FAQ Search: Navigate to the FAQ section and locate the explanation for the question 'What is blockchain?'.